

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 27)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			
Use of materials			
Nature of wastes			
Temperature of processes			

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

01 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

.....

.....

.....

.....

01 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

.....

.....

.....

0 1 . 4

Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

10

0 2 . 1

State what is meant by **hybrid vigour**.

[1 mark]

0 2 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

0 2 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

03

The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

03

1

Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03

2

Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03

3

Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

04 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

04 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

04 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

10

0 5 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

Ozone depletion

Tourism

Future mineral exploitation

0 5 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

10

06 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species

.....

.....

.....

Ecological niche

.....

.....

.....

Community

.....

.....

.....

Biome

.....

.....

.....

06 . 2 Explain the difference between an ecosystem and a biome.

[2 marks]

.....

.....

.....

.....

10

07 . 1

Which of the following best describes a **representative** sample? Tick one box.**[1 mark]**

A A sample taken from the most convenient location

☐

B A sub-sample that accurately reflects the complete data set

☐

C The largest sample that can be collected

☐

D A sample that supports the hypothesis

☐

07 . 2

Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

.....

.....

.....

.....

07 . 3

Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

.....

.....

.....

.....

5

08 . 1

Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

.....

.....

.....

Endemic species

.....

.....

.....

Keystone species

.....

.....

.....

Flagship species

.....

.....

.....

08 . 2

Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

.....

.....

.....

.....

10

0 9

Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 9**. 1**

Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9**. 2**

Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9**. 3**

Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

.....

.....

09 . **4** Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

10

1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

10.1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer	%
--------	---

10.2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

1 0 . 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

.....

.....

.....

.....

.....

.....

1 0 . 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

Extra space

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

1 1 . 2

Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

END OF QUESTIONS